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A systematic review on CNN-YOLO techniques for face and weapon detection in crime prevention

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Abstract
The increase in criminal activities has led to an increased focus on crime forecasting and proactive measures using advanced technologies to detect crime activity before the crime has to be committed or suspected. A major challenge is to prevent such crimes and protect people from criminals with the help of deep learning technology with advanced neural networks, computer vision, classifiers, and methodologies such as you only look once, Faster RCNN, convolution neural networks, and GANs, which are employed to forecast crime activities while preventing criminal activities before they happen. Crime analysis revealed two key elements like facial recognition and weapon detection and these technologies are beneficial in identifying criminals and determining their exact situation. Currently, governments and private institutions use closed-circuit television (CCTV) based surveillance to prevent and monitor crime. These techniques provide high performance and accuracy in recognizing criminals. This review aims to highlight the current state of these technologies, their impact on crime analysis, and potential future efforts related to detection under difficult conditions, minimizing false alarms, and integrating detection models into real-time, scalable, and privacy-preserving surveillance systems developments to improve security and law enforcement practices.

Keywords Criminal analysis, Deep learning, Deep convolution neural network, Faster region-based convolution neural network, Generative adversarial network, YOLO

1 Introduction

Deep learning technologies have emerged as powerful tools to include into surveillance-based crime prevention. These approaches have multiple advantages in their application, but none more so than incorporating Convolutional Neural Networks (CNNs) with (formerly) real-time object detection model like You Only Look Once (YOLO). CNNs are excellent at extracting features from images and videos in a more heuristic manner, while YOLO is highly effective for a one-pass, accurate, and speedy localization and classification of the behaviour of interest. The pair together really provide the ability to detect in real-time and, thus efficiently and accurately detect face, weapons, and suspicious

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behaviors discovered in live feeds of surveillance. This technology thus gives an instantaneous response that law enforcement systems reap the benefits of, allowing them to close the loop from detection to response immediately and engaging more effectively than other traditional models whereby the post-event analysis provides only perspective to what occurred, not actionable intelligence to response.

Detecting crime activities is essential because it highlights the pressing need for innovative solutions to improve public safety measures. CCTV surveillance and detection technologies help anticipate and prevent crimes, leading to safer regions and more dynamic policing. Visible surveillance acts as a barrier, whereas high-quality footage serves as evidence of carping, strengthening the judicial process. Effective crime prevention builds public faith and helps in developing informed policies to address conflicts more effectively. Crimes can be differentiated into violent crime activities and nonviolent crime activities. In this study [1], the crime rate report published by the National Crime Records Bureau (NCRB), under the Indian government, records all types of crimes. Every year, the state government submits the reports to the State Crime Records Bureau (SCRB), which collects the data from the District Records Bureau (DCRB) and submits all the data to the NCRB for its annual reporting. In India, thieving accounts for the majority of crimes committed, followed by robbery and assault. Generally, the crime rate is substantially greater in metropolitan areas than in rural areas. The National Crime Records Bureau (NCRB) released a 2023 report indicating that the reported crime rate in India was 445.9 per 100,000 individuals. The overall crime incidence in India increased by 120% from 2013 to 2022; In 2022, 48% of all cognizable crimes were reported in the 5 states Uttar Pradesh, Maharashtra, Gujarat, Madhya Pradesh, and Tamil Nadu. In the Indian government, the NCRB published a statistics crime rate report from 2014 to 2022 as shown in the graph.

In Fig. 1 As a population grows, crime rates tend to increase proportionally due to higher populations. In 2014, the crime rate was reported to be 4,571,663 and drastically increased until 2020. The crime rate was reported to be 6,601,285 in 2021; the crime rate was reported to be 6,096,310; and in 2022, the crime rate was reported to be 5,824,946. This paper discusses crime activities and focuses on two types. Crimes against persons

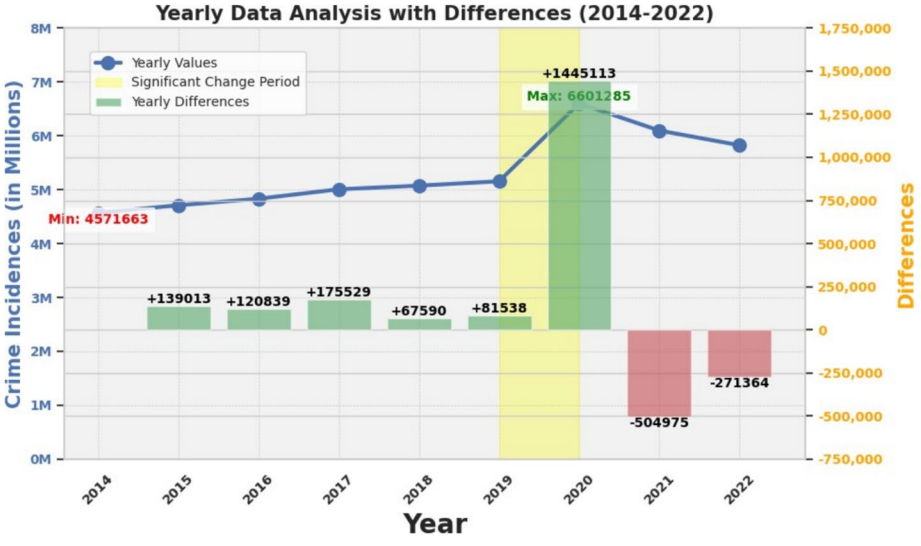


Fig. 1 Yearly crime report analysis https://www.themirrority.com/data/crimes

such as assault, violence, robbery and crimes against property such as forgery and burglary. The analysis of crime focuses on 2 key elements, facial recognition and weapon detection via deep learning methods. Facial recognition predicts the individuality of criminal activities via facial expressions, weapon detection, identification of firearms and dangerous objects, and violence detection to analyze behaviors and body movements. Additionally, facial recognition and weapon detection are important. Humans can easily predict a large range of facial expressions without parallel skills when faced with face-to-face computer machines. Furthermore, 55% of the emotions and feelings that people experience are elicited by facial recognition [2]. Automated facial recognition (AFR) technology uses algorithms to detect a face, create a digital representation of it, and compare it to other images to determine how similar the two main things are to AFR technology. Verification means that one can confirm one's identity by comparing a single image, such as a passport photo, to another image, such as a border control gate photo.

The violent crime types include assault and robberies, and the nonviolent crime types include morality, White collar crime, and organized and cybercrime. Violation types impact people and communities in several ways. According to Fig. 2, victims of violent crimes suffer not only from physical injury but also from long-term trauma and psychological harm. Property crimes suffer financial loss and emotional distress, which are worsened by the raid of their private space. White-collar crimes suffer economic loss, affecting their living and guarantee. The historical context of crime shapes social perceptions, cultural biases, and legal frameworks [1]. A multifaceted approach to crime prevention requires law enforcement partnerships, legitimate action, public involvement, and social mediation to address the multiple impacts on personnel and society. In this study, biometric-based technologies such as genetic biometrics were confirmed and identified by examining human body characteristics, such as fingerprints, pupils, and veins. Face algorithms of local binary patterns were utilized to reduce the impact of light exposure, which affects the accuracy of the system. Identification means that a database of criminals, such as CCTV images, can be searched to determine if they match the target image. This can be done by one person, an algorithm, or a combination of one

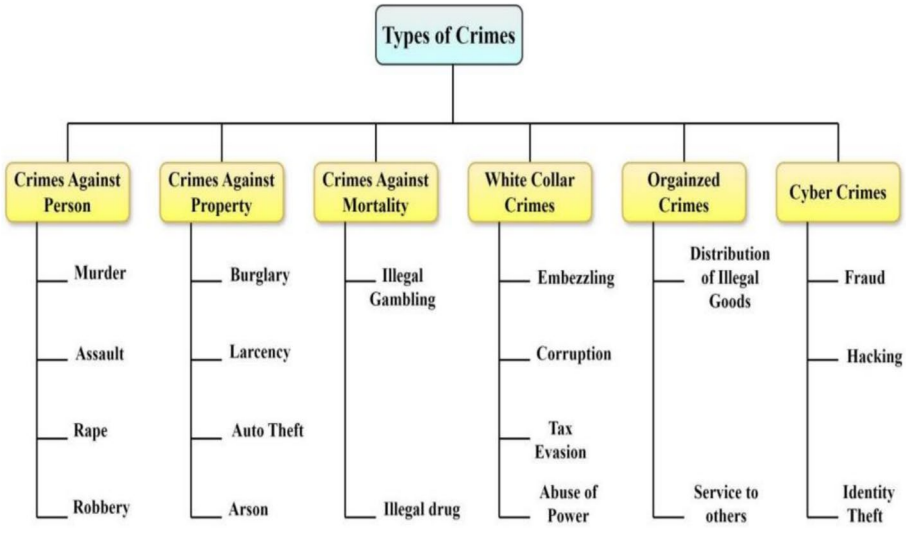


Fig. 2 Type of crimes

person and an algorithm. We provide more detail about the different types of algorithms and their operational uses.

The Integrated System for Facial Expression Recognition (ISFER) did automatic analysis of human facial expressions by automatically recognizing faces and sorting facial expressions into categories based on their emotional content [3]. Karolina Nurzynska et al. [4] also came up with a manual preprocessing system for recognizing facial expressions. Researchers have focused on the problems of detecting action units in neutral settings, pose variation, rigid head movements, and lip movements for automated facial image analysis and synthesis [5]. Age-based models have a lot of uses. For example, police may easily discover missing people and catch the most wanted offenders with image processing techniques for security purposes [6]. Another factor is weapon detection during the crime. This is a safe haven activity that employs cutting-edge open-source deep learning algorithms to figure out which weapons are harmful from CCTV photos [7]. The UK has about 4.5 million cameras, whereas Sweden has about 50,000 cameras installed per year. These cameras come from different countries throughout the world. By putting up only 450 cameras in the city of Poznan, the Polish government was able to cut drug cases by 60% and street fights by 40%. However, the problem of overloading the human operator is the use of automated picture understanding algorithms [8]. The automatic detection of firearms as guns involves a combination of color-based segmentation and interest point detection [9]. Deep learning methods like recurrent neural networks and convolutions seem to be able to help forecast crime. To correctly guess crime patterns in a specific city, these algorithms are trained using crime data that has a spatial or temporal component. For instance, algorithms have been used to look at criminal data such the time, place, and type of crime. We utilize these data to develop predictive models that can help us find high-risk areas and guess when crimes will happen in the future.

The deep learning applications for criminal prediction includes computer vision and video analysis. Technology has been used to observe and classify crimes such as mischief, theft, and assault in analysing video clips from surveillance cameras. Advanced deep learning models can be used in combination with aerial technologies or drones providing engagement opportunities and surveillance intelligence that could actively monitor criminal activity. These algorithms can also be used to analyze sources for criminal activity across various authorities such as crime data reports, social platforms and police source reports to create a holistic view of criminality and activity using automated sources. Deep learning would assist in identifying and acknowledging crimes in real time therefore making it a viable resource for crime intervention and combating crime. As crime increases there is already demands to extend smart surveillance systems to limit the traditionally limited post-event analysis and it has become evident, there are limited real time responses and even more limited accuracies once complex scenarios such as occlusion, low resolution or high pedestrian count conditions occur. This work will examine and integrate testing deep learning models (CNN, YOLO, and Faster RCNN) for pro-active crime prevention real-time face and weapon detecting based on implemented CCTV images.

The challenges of traditional systems include the following:

- Sketches created by humans may be vague and imprecise.
- The crude drawings of a target could be monochrome or black and white.

- Color dependent characteristics are not important.
- Some aspects are modifiable by the offender or target. For example, hair color, and haircut.
- If the target face must locate a match without the aid of a database, human facial recognition will take longer.
- Struggle for accurate detection of a wide variety of weapons, especially nonmetallic or those hidden under clothing.

Systems produce false alarms, causing unwanted panic and operational interference. This difficulty is due to overlapping objects and people. Many struggle to adapt swiftly to new types of weapon methods used by criminals. The traditional system of facial recognition and weapon detection systems struggles with indistinct, monochrome sketches and modifiable elements, leading to stagnant human recognition, false alarms, and difficulty in adapting to new criminal methods and complex situations. These systems typically require human operators to manipulate review alerts and images, which can be time-consuming and increase the chances of human mistakes.

Although there has been some progress in incorporating deep learning into surveillance systems, this research contributes novel knowledge regarding how YOLO, Faster R-CNN, and CNN-based models can be combined in real-time to assist more efficiently in proactive crime prevention and public safety. Although some surveillance systems may incorporate facial recognition and weapon detection functions, these functions are typically done post-event. The valuable contribution of this research comes from performing real-time detections as they occur. The use of YOLO and Faster R-CNN models that facilitate real-time detection and tracking in conjunction with CNN-based models provides a platform from which real-time detection capabilities can be improved over prior methods that rely on imperfect human descriptions, human appearance traits that can change and limitations of existing surveillance systems to detect hidden weapons. The proposed system would algorithmically collect and maintain biometric identification data and the identification method of using real-time video feeds from the surveillance data/video previously mentioned would close the gap between detection and action—allowing for law enforcement/police officials to take a more proactive response to incidents that may be occurring. The discussion is positioned with consideration of national crime statistics, linking the increasing crime rate with the rising need for scalable, intelligent, and privacy-conscientious surveillance models. The research study contributes novel progress as a not just a technological contribution, but as a functional advancement toward improving public safety infrastructure.

2 Literature survey

A survey of various uses of deep learning (DL) techniques in crime analysis covers the methods, algorithms, and research areas identified. Many DL models, such as Faster RCNN, YOLO, and Mask RCNN are used for tasks such as object detection, anti-theft system development, and fraud detection of Wi-Fi signals. The study also discussed the reasons for the number of guns and weapons and the difficulty in recognizing the face through the eyes. Many studies have focused on violence detection, including the automatic identification of violent videos to protect children. The effectiveness of DL in crime prevention is demonstrated through applications such as searching for masks and weapons and using email alerts forfeits. Recent advances such as YOLOv4 have

improved vehicle target detection and innovative approaches, such as deep search. Overall, Table 1 demonstrates the utility and potential of DL models for improving various aspects of crime analysis and investigations.

Pavithra et al. [11], analyzed suspicious activities through CCTV surveillance videos, which play a major role in residential area security through the advancement of a surveillance video structure to identify any type of abnormal activity, especially robbery activity. This system helps with the effective use of deep learning techniques, such as the YOLO algorithm, to detect robbery via techniques such as object detection and identifying the actions needed to prevent robbery. YOLO was used to determine whether the activity was suspicious, and the YOLO architecture was used to construct the CNN model. Lian Rebello [20] analyzed fraudsters who utilized deep fake techniques and easily identified the differences between real and deep-fake videos. LSTM for temporal sequences and CNN for feature extraction plays a major role in detecting the activities

Table 1 Criminal type analysis via deep learning techniques

S.No	Refs.	Methods	Advantages	Limitations
1	Rajdeep Chatterjee [10]	Faster RCNN, EfficientDet	The automated detection system reduces human intervention	This method does not cover variations in weapons and still has the possibility of incorrect detection
2	Pavithra [11]	CNN, YOLO	Easily identify deep fake videos and an accuracy of 81%	There is a lack of full validation in real-world scenarios
3	Apar Jaggi [12]	Vision-Wi-Fi cross-model, Mask R-CNN	Using real-time monitoring, reducing human error, and attaining an accuracy of 98% Violence was detected in real- time scenarios and the data were analyzed quickly	Generate the false alarm rate and challenges of real time deployment
4	Tasnim Sakib [13]	"VGG-16, Inception V- 3, ResNet50 MobileNet V-2 Xception"	Violence was detected in real-time scenarios and the data were analyzed quickly	High computational cost, false negatives, and positives
5	Nikkath Bushra [14]	Multiresolution image mosaic(MRIM) Haar Cascades Convolutional neural network(CNN) Google Object	Automated surveillance, real-time action recognition	Data collection details and limited scope
6	Khalid Raihan [15]	Convolutional neural network(CNN) Google Object	Easy to handle various input types and enhanced criminal investigations	Complexity of GANs and limited datasets
7	Xizhi Hu [16]	YOLO	Reduces false alarms Real-time detection	Does not discuss how well the model generalizes to different environmental contexts, such as varying lighting conditions or occlusions
8	Yesun Utomo [17]	Masked Face recognition (MFR)	An innovative approach leads to real-time application and achieves a high accuracy rate	The datasets are not representative of real-world scenarios, and the model's effectiveness could be complicated
9	Abdul Rehman [18]	CNN, SMTP, DCGAN	By reporting ongoing illegal activity, the system helps prevent future robberies from occurring	Poor lighting, camera angles, and image quality can affect detection accuracy and the risk of false positives
10	Swati Jagtap [19]	CNN, Transfer Learning	Advanced detection is more efficient and easier and reduces human intervention	Incorrect alerts or missed detections

of fraudsters. MesoNet manages the noise produced by low-quality video processing and deep fakes jeopardize facial recognition and online content. Fake news via social media platforms such as Facebook and Instagram is used to analyze news posts, and if news is fake, it automatically alerts the user automatically. In Poloju Akshaya [21], CCTV surveillance plays a vital role in observing activities such as robbery and threats in public places. This surveillance requires human observation to avoid these problems and implement an automatic weapon detection system. Using these techniques, delicate learning methods can be used to easily identify weapons such as pistols, guns, and knives.

Shah et al. [22] surveyed street crimes and suggested that these crimes be controlled by installed X-rays or employed police to avoid the high cost and high amount of manual human activities needed to implement the weapon detection system using CNN architectures such as a mixture of Gaussian's, polarized signal based techniques, and the Haar cascade procedure. Khalid Raihan Talha et al. [15] suggested preventing violent acts in real-time while monitoring systems are capable of detecting violence very swiftly. The system is implemented to detect the speed of people's movements hand and leg movements and their distances from one person to another person. Suppose these persons are close to each other while the system alerts the message to the responsible channel. A CNN-BiLSTM based violence detection system ensures that the public has a secure place for peace-loving people.

All crime activities to be detected by the object detection algorithms of YOLO, Faster RCNN, and scalable and efficient object detection (EfficientDet) object detection algorithms are commonly used in real-time environments and achieve high accuracy in facial recognition and weapon detection. Widely used classification models, such as CNN's, enable feature extraction and classification tasks and help enhance detection accuracy. This table shows research gaps concentrated on improving detection accuracy, reducing false positives, and addressing conditions such as less bright or masked faces. These methodologies are applied in various domains such as smart city surveillance, anti-theft systems, vehicle detection, violence detection, and facial recognition. Several deep learning models can be applied to different datasets for tasks such as object detection, facial detection, and weapon detection. Shanthi P [23], an integrated framework that fuses weapon detection and human activity recognition in surveillance footage. It fuses object recognition and activity understanding to jointly detect weapons and suspicious activities. The comprehensive system improves situational awareness, facilitating proactive crime prevention and real-time threat response.. Initial testing on standard biomedical imaging datasets has shown promising results, suggesting that this method could lead to more efficient screening procedures and personalized treatment strategies for oral cancer [24]. Greylag Goose Optimization (GGO) algorithm, a novel nature-inspired metaheuristic optimizer inspired by the behavior of Greylag Geese [25]. A novel hybrid metaheuristic optimization algorithm, GWDTO, which combines Grey Wolf Optimization (GWO) and Dipper Throated Optimization (DTO) for feature selection in Network Intrusion Detection Systems (IDS) [26].

Abdul Rehman et al. [18] suggested that knives and assault are frequently used weapons for various terror incidents and lack security checks; consequently, CCTV cameras are difficult to use for human monitoring. To overcome these issues, automatic weapon detection systems for detecting firearms and knives have been implemented via models such as YOLOv5 and Google Colab Pro. To select the hyper parameter file that provides

a basis for evolving minimal hyper parameters, such as the learning rate, momentum, weight decay, box loss, and object loss, IOU training threshold augmentation models such as gamma, flip, translate, and scale are used. Three challenges face this paper: 1. Image 2. Weapon 3. Color like lighting conditions. Using deep learning techniques such as the YOLO model achieves real-time weapon detection and produces very few false alarms. Apar Jaggi [12] proposed a system that easily identifies any cause of antisocial or abnormal activities or assault by working on models such as transfer learning via a deep convolutional network (DCN), a state-of-the-art CNN model that uses recent object detection techniques such as YOLOv7. The model was trained via the YOLO algorithm. Poornima et al. [27] presented the sketch image classification method in the CNN model for the recognition of facial photos, which is a large task network, and the facial identification system takes color and sketch images as inputs.

Combining YOLOv8 with FMR-CNN utilizes the robust feature extraction and region-based precision of FMR-CNN and the real-time speed and efficacy of YOLOv8. This blended methodology enhances both detection and processing speeds, which in turn makes weapon detection within surveillance systems more efficient and accurate. [28]. The proposed system, which includes image preprocessing, a fusion model, and an optimized feature selection algorithm, demonstrates superior accuracy on benchmark datasets, offering a promising solution for monitoring suspicious activities [29]. The real-world problem and a practical need for automated systems. The proliferation of surveillance infrastructure creates a massive amount of data that is impossible for humans to monitor continuously [30]. AO-DETR, an anti-overlapping detection transformer designed to improve the accuracy of prohibited item detection in X-ray images, particularly addressing challenges posed by overlapping objects [31].

To address two steps 1. Facial recognition 2. Synthesis to produce better results via networks such as GANs and CNN. Nikkath Bushra et al. [14] presented a technique to categorize the activities such as normal and abnormal activities, without attempting to guess that a situation occurred in a certain area as abnormal via techniques such as the Mobile Net, ResNet50, and CNN algorithms and the SMTP library to easily detect the face mask of burglars and object details and send an email alert to the respective department. Xinyue Fang et al. [32] presented WiSil, a novel video forgery detection system that is designed to secure video surveillance and forensic systems by leveraging WiFi signals to detect and manipulate video frames in real-time. Although it focuses on live video forgery detection, it is video/camera-based surveillance systems have been applied in a wide spectrum of applications, such as retail, banking and logistics, and generally Internet of Things (IoT) devices and the emergence of deep-fake techniques emphasizes the discerning of video surveillance systems under such attacks. WiSil achieves 95% accuracy in detecting tampered frames and is robust against environment and person variations. Table 2 presents the classification model framework for different types of crimes via several deep-learning models. This table is split into two major categories of crimes: 1. Crimes against person 2. crimes against property. Crimes against person crimes include robbery [11, 20], violence [13, 15, 21, 22] and assault [12, 18, 27] and crimes against property as burglary [17] and forgery [32].

The CNN model is commonly used for image and video recognition and the other models can support large amounts of sequence data processing systems. Object detection models use algorithms such as YOLO, SSD, Faster RCNN, and efficient detection.

Table 2 Classification and object detection model

Classification Model	Crimes Against Person		Crimes Against Property		
	Robbery	Assault	Violence	Burglary	Forgery
	[11, 20]	[10, 15, 17, 21]	[14, 22, 27]	[19]	[32]
CNN	✓	✓	✓	✓	✓
LSTM		✓		✓	
RNN		✓	✓		
GAN	✓			✓	
DCNN			✓		✓
Object detection Model					
YOLO	✓	✓	✓	✓	
SSD	✓	✓	✓	✓	
R-CNN	✓	✓			✓
FASTER R-CNN	✓	✓	✓		
EfficientDet		✓			

This complete review serves as an example for understanding the potential applications of multiple techniques in crime classification and detection.

3 Methodology

A lot of research has been done on deep learning models for crime detection, but most of it just looks at facial recognition or weapon detection on its own. There aren't any integrated real-time solutions. Numerous models undergo evaluation in optimal conditions but fail to transfer to real-world CCTV settings characterized by obstruction, insufficient lighting, and high crowd density. Additionally, there is limited research comparing CNN, YOLO, and Faster R-CNN in the context of dual-task detection. This study tackles these deficiencies through a systematic review and a model evaluation framework. Crime aspects are reviewed as two key elements: first, they include criminal facial recognition and weapons. Detection uses advanced facial recognition technology to identify people who are willing to engage in criminal activity. This process uses algorithms and large datasets and aims to match facial features to known criminal databases, helping the police to foresight and prevent crime and target surveillance. While significant progress has been made in terms of accuracy, challenges remain in ensuring proper use, addressing unfairness, and protecting privacy. As tools for modern policing, face and weapon recognition of criminals has generated debate about its effectiveness, possible social effects, and the need for strict regulations to balance security and civil liberties. In this paper, we discuss the Neural Networks and You Only Look Once models.

3.1 Facial recognition

Most CNN applications focus on image data, such as object detection, optical character recognition, image classification, and facial recognition. This paper discusses the use of facial recognition technology for identifying criminal faces. The criminal record contains personal data and photographs.

To identify a criminal, eye-ness data about the person are needed; these data can be obtained through biometric technologies such as fingerprints, DNA, and irises and through the optimization of visual salience [33], the technique used in scientific visualization to increase the region of interest within volumetric data. In Fig. 3 Facial recognition technology is simpler than these other methods because the face is the primary focus of social interaction. In addition, the information needed to create a facial

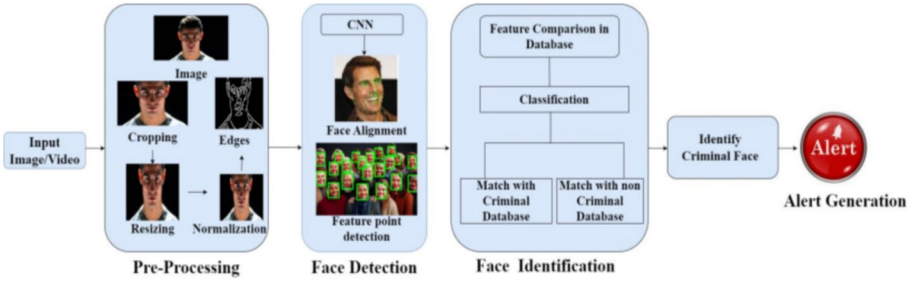


Fig. 3 Face recognition

recognition model can be easily obtained from criminal records, videos, and other sources. The constructive solid geometry (CSG) approach applies the local coherence of facial space labels to accelerate the facial membership classification process [34]. Facial recognition has also been used in other areas. For example, payment companies, such as MasterCard and Alipay, have developed systems where users pay for selfie. Facial unlock is a feature of many smartphones. The deep convolution neural network (DCNN) algorithm for analyzing and verifying facial expressions in real-time [14] plays an important role in criminal investigations, especially in the analysis and interpretation of visual data.

3.1.1 Image and video analysis

Neural networks are used to analyze images and videos from surveillance cameras, crime scenes, or other sources. They can identify and classify objects, persons, or activities related to criminal investigations. In this study, Sreenu et al. [35] used a convolutional restricted Boltzmann machine (CRBM) and a restricted Boltzmann machine (RBM), which are both techniques used in image and video analysis, to concentrate on image patches and compare entire frames to address the problem of object detection in crowded scenes. The SVM classifiers take the produced elements as inputs for facial detection and a custom VGG16 model for emotion identification. A robust representation of body motion via 3-D joint coordinates generated from any motion capture system [36] focused on an image and video analysis easily predicts scenes that could be violent or nonviolent in a particular location. Video emotion prediction is performed by producing spatiotemporal elements through additional encoding temporal information [37], and facial expression video GAN(FEVGAN) analysis of the image videos of 86 people are performed on facial expressions such as happiness, sadness, surprise, anger, disgust, fear, and neutrals to train and produce the output of what they have given facial expressions [2].

Convolutional restricted Boltzmann machine (CRBM)

CRBMs expand by incorporating convolutional layers. This makes them suitable for tasks involving spatial hierarchies such as image patches to be more concentrated on the energy function in Eq. (1) and contrastive divergence in Eq. (2).

Energy function

V visible layer h hidden layer wi, j = weight matrix ai and bi are biases.

$$E(v,h) = -\sum a_i v_i - \sum b_j h_j - \sum v_i w_{i,j} h_j \tag{1}$$

Contrastive Divergence (CD)

$$w = \varepsilon\left((v_h)\,data - (v_h)\,model\right) \tag{2}$$

E – learning rate.

(v_h) is the expectation from the data distribution.

The (v_h) model is from the model distribution.

Restricted Boltzmann Machine (RBM)

It is utilized for unsupervised learning tasks. These methods are particularly useful for feature extraction and dimensional reduction as in Eq. (3) and (4).

Probability distribution

$$p(v,h) = 1/z\ \exp(-E(v,h)) \tag{3}$$

Where Z is the partition function:

$$z = \sum_{v,h} \exp(-E\,(v,h)) \tag{4}$$

3.1.2 Facial detection

Facial detection tasks help law enforcement agencies match faces in images or videos with known individuals from databases. In facial recognition, initially an input must be detected and verified. The input images can be recorded as video or real-time video. The regularized broad learning system (RBLS) approach can synthesize high-quality facial sketches efficiently and quickly [36]. The next step for accessing faces in images or videos is to perform accurate detection once the classifier is trained well. As presented in ref. [38], Haar feature-based cascade classifiers and CNN utilized a classifier, as the Haar cascade focused on the frontal face and improved real-time performance. Many approaches to coding image region content include local binary patterns, uniform patterns, and rotation invariant LBPs successful detection and recognition of the face of a person in real-time or recorded video frames.

Haar feature-based classifiers

These features are based on simple rectangular patterns (Haar features) that capture the differences in intensity between adjacent regions. The cascade structure allows for the rapid elimination of non face regions by using a series of increasingly complex classifiers. The tool is designed to predict frontal faces effectively, making it suitable for scenarios where faces are mostly aligned and facing the camera. Haar cascaded images are computationally efficient because they use internal images, which allows for rapid computation of Haar features across different regions of the image.

$$\begin{aligned} Haar\ feature\ &= Sum\ of\ pixels\ in\ the\ white\ region \\ &\quad - Sum\ of\ pixels\ in\ the\ black\ region \end{aligned}$$

Local binary patterns (LBP)

Binary pattern calculations require comparing the intensity of each pixel with that of its neighbors and encoding these comparisons into a binary pattern in Eq. (5). This pattern captures local texture information, which is useful for distinguishing different faces. Rotation invariance extensions of LBPs, such as rotation invariant LBPs, improve the recognition performance by accounting for rotations in the image, increasing the versatility of the method.

$$LBP = \sum_{p-1}^{i=0} s(I_i - I_c).2i \tag{5}$$

where P $_i$s the number of surrounding pixels, Ii is the intensity of the Ii-th pixel, I_c is the central pixel intensity, and s is the step function that outputs 1 if IiIc and 0 otherwise.

Identifying characters

Specified characters are analyzed with the help of patterns in visual data, and investigators identify potential trends, correlations, or anomalies related to criminal activity. Image enhancement: These meshes contribute to image enhancement, improving the clarity and quality of the visual evidence that is essential for research. A broad learning system provides visual analytic tools such as radar figures, iceberg visual imagery, and heat maps to enhance the understanding of player performance dynamics in matches; however, these tools typically fail to correct predictions in real applications where only some untruth samples created by unseen forgery methods are available. To avoid this problem [39], a method of boosting performance, such as unseen techniques, was implemented termed meta-fakeness which requires only a fake sample. Another way to move the head and body is to monitor criminal cases [40], identifying and detecting head movements, such as forward, leftward, and rightward movements, by using the distance of the eye location methods implemented in facial edge detection to obtain high accuracy. Position-based dynamics (PBD) methods incorporate both coarse volumetric and detailed collision handling through a hybrid Eulerian /Lagrangian framework [41].

Contrast adjustment

The contrast of an image can be mathematically adjusted as follows in Eq. (6):

$$I_{enhanced(x,y)} = \propto I\,(x, y) + \beta \tag{6}$$

Where I (x, y) is the original pixel value at position (x,y), α is the contrast factor β is the brightness offset.

Meta fake method

The boosting performance approach to enhancing performance via fake samples can be represented as shown in eq. (7)

$$L_{boosted} = L_{real} + L_{fake} \tag{7}$$

$L_{boosted}$ is the boosted loss function, L_{real} is the loss on real samples, L_{fake} is the loss on fake samples, and λ is a weighting factor.

Head and body movement detection

The eye location distance indicates the distance between eye locations for movement detection and can be computed via the following formula as in eq. (8):

$$\mathrm{d} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \tag{8}$$

where (x_1,y_1) and (x_2,y_2) are the coordinates of the eyes in two frames.

Facial edge detection

The edge detection Sobel operator can be used for edge detection and is represented by in eq. (9)

$$G_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix} * I \quad G_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{bmatrix} * I \tag{9}$$

where G_x and G_y represent the gradients in the x and y directions, respectively and * denotes the convolution operation with an image I.

START
Step 1: Image Acquisition of Dataset
Images = Load Dataset ("Path
Directory")**Step 2:** Feature Extraction
Initialize Model
model = initialize ()
Step 3: Image Classification
Predict whether the image contains a criminal facial or noncriminal
facial Predictions = model. predict (images)
Step 4: Match the criminal
imagefor prediction
it is a criminal image(prediction):
Result = "Criminal face"
else:
Result = "Non-Criminal"
Print (result)
STOP

Algorithm 1 Pseudo code for Criminal Face Detection

The provided pseudo code Fig. 4, framework process for detecting criminal images via a convolution neural network (CNN). The process starts with the acquisition of a dataset containing images, which are subsequently preprocessed for feature extraction. A CNN

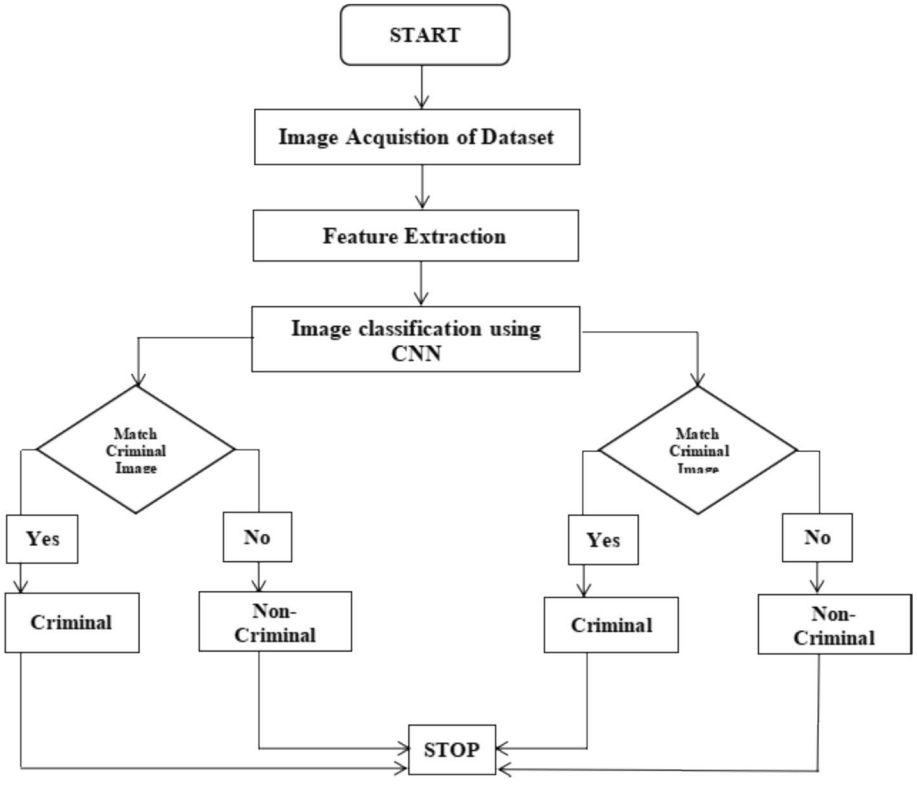


Fig. 4 Criminal facial detection

model is initialized to perform this extraction in parallel. The images are categorized via the model to determine if they match any criminal images each and every prediction is evaluated. If the image matches a criminal profile, it is labeled "Criminal"; otherwise, it is labeled "Non-Criminal". These processes involve iterating through each image and outputting the appropriate classification before concluding. This structured approach ensures efficient and accurate image classification for identifying potential criminal content.

Flowchart

Using YOLO and Faster R-CNN enables us to detect faces and weapons at the same time from live feeds of surveillance video. YOLO's unified architecture allows us to quickly identify multiple objects and Faster R-CNN's RPN can identify more accurate candidate regions to allow it to accurately identify critical objects of interest like faces and weapons to ensure quick and accurate monitoring of criminal activity.

3.2 Weapon detection

Weapon detection plays a vital role in public safety, as it allows for prior information and identification of threats and quick intervention by departments such as police and law enforcement. Analyzing weapon detection data can increase system accuracy, minimize false alarms, and ensure reliable weapon detection. Neural network technology is important for high-risk settings such as schools, colleges, airports, temples, and public gatherings, where the impact of a weapon. In this survey, weapon detection via these deep-learning technologies was investigated.

3.2.1 Object identification

Object detection is utilized to detect and print specific objects such as weapons or suspicious objects to help in crime scene analysis. Weapon detection is an essential factor for correctly identifying and discovering potential threats in real-time and ensuring public safety. SparseVoxNet [42] integrates volumetric data with surface normal and curvatures to optimize 3-D object recognition, detect objects within the image or video frame, and produce bounding boxes around latent objects of interest. In this study, Bo Zhang [40], applied nonuniform imaging, specifically for LiDAR systems, which are used for automated detection. The Mask R-CNN model helps object detection identify dangerous objects indoors during suspicious activities taken in any place [13]. Figure 5 describes the stages of weapon detection in an image and video, it takes as input, accompanied by preprocessing, which includes image scaling and noise reduction. Objects are detected, and weapon classification classifies these detected objects. Environmental analysis to

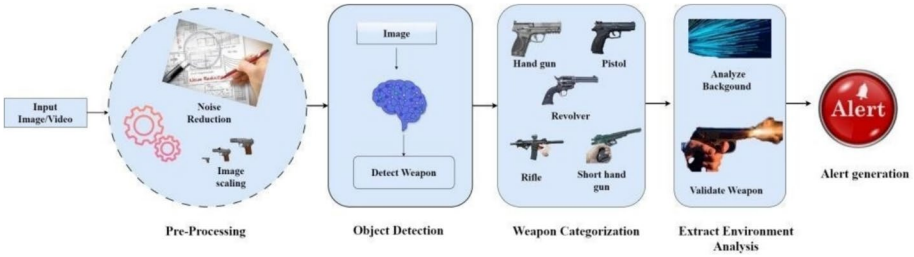


Fig. 5 Weapon detection

validate the situation of the particular place of actually detected weapons at the last alert generation creates alerts for identified threats.

Mask R-CNN model

Faster R-CNN was extended to create a Mask RCNN by adding a branch for predicting segmentation masks in each region of interest (RoI) in parallel with the previous branch for classification and bounding box regression.

Region proposal network (RPN)

The RPN generates region proposals, which are possible bounding boxes around objects. The loss function for the RPN combines classification and regression losses in eq. (10).

$$L_{RPN} = \frac{1}{NCLS}\sum(p_i, p_i^*) + \lambda\frac{1}{Nreg}\sum p_i^* L_{reg}(t_i, t_i^*) \tag{10}$$

P_i is the anchor i predicted probability; p^*_i true positive identification; N_{reg} t_i, t_i^* anticipated and ground truth box coordinates; λ balances 2 loss terms.

RoI alignment The mask RCNN involves RoI alignment instead of RoI pooling to avoid quantization errors. It interpolates the features from the feature map more precisely in eq. (11).

$$ROI_{Align}(x, y) = \sum w_{i,j} f(x_i, y_j) \tag{11}$$

where ($w_{i,j}$)are the interpolation weights and $f(x_i, y_j)$ are the feature map values.

Start
Step 1: Data collection X = x_i
Step 2: *Data preprocessing* $X_{pre} = f_{pre(X)}$
Step 3: *Data augmentation* $X_{aug} = f_{aug}(X_{pre})$
Step 4: *Model Selection* $Y = f(X_{aug}; \theta)$
Step 5: *Model training and evaluation* $Y = f(X_{aug}; \theta)$ θ = *Model parameter*
Step 6: *Model Prediction* $y_pred = f(x_input; \theta)$
Step 7: *if* $y_pred >$ *Threshold Result* (*weapon detection*)
else :
result (*non weapon detection*)
Print(*result*)
Stop

Algorithm 2 Pseudo code for Criminal Face Detection

Flowchart

Bounding box regression

Adjusts the bounding boxes to give a good fit to the predicted objects in eq. (12)

$$\begin{aligned} t_x &= \frac{x - x_a}{w_a} \\ t_y &= (y - y_a)/h_a \\ t_w &= \log(w/w_a) \\ t_h &= \log(h/h_a) \end{aligned} \tag{12}$$

where x,y,w, h are the center coordinates, width and height of x_a, y_a,w_a , and h_a are the anchor box parameters

3.2.2 Weapon categorization

The purpose of weapon classification is to clearly identify types of weapons, such as guns, knives, and pistols, and to qualify effective warning assessments and connected security responses.

To train the neural network model via the preprocessed datasets, which demand that the images be fed into the CNN to extract relevant features from the layers and classify the images into different weapon categories, a novel efficient [43], and robust Boolean evaluation approach based on hybrid representation was proposed in Fig. 6. In this study [18], followed a structured approach that involved selecting appropriate architectures such as VGG-16, ResNet-50, and ResNet-101 to achieve better performance.

3.2.3 Environmental analysis

With the help of interpreting complicated scenes by detecting and understanding elements such as vehicles and activities in jammed or dynamic environments, transformer-based methods [44] utilize predefined pose templates and a transformer architecture to generate human poses that are contextually relevant to specific scenes. Crime activities involve examining the physical and social environments where crimes occur to identify patterns and elements that influence crime behaviors. To collect data on crime incidents

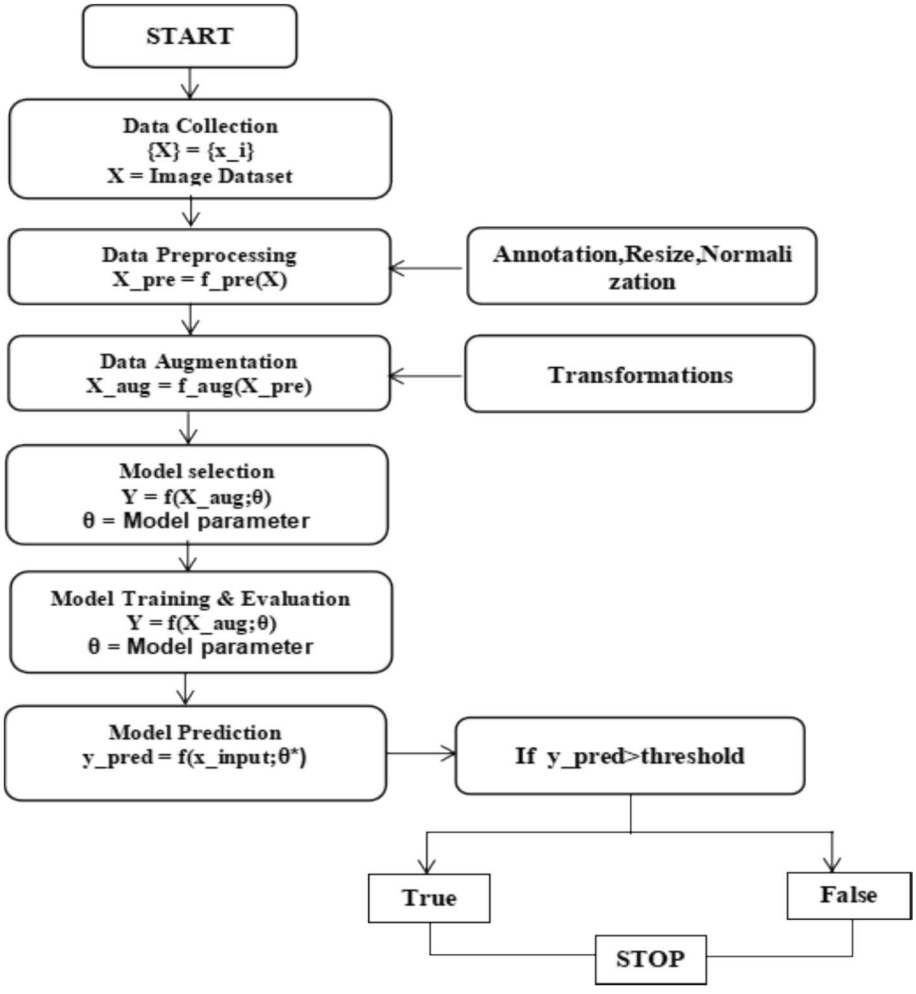


Fig. 6 Flowchart of weapon detection

such as location, time, type of crime, census-taking information, and spatial and temporal analysis. Once the environmental scenes in a particular location are identified, there may be fewer opportunities for crimes. Let us illustrate this phenomenon via crime hot spot analysis. Here, Z is the Z score.

$$\mathrm{Z} = \frac{\sum_i (x_i - y)}{\frac{\sqrt{\sum_i (x_i - y)^2}}{n}} (13)$$

x_i represents the individual crime incident locations. Y is the mean location. N is the number of incidents.

3.3 Neural network approaches

In general, deep learning processes massive amounts of data via contemporary algorithms, improving understanding and decision-making. Using techniques such as facial recognition, pattern recognition, and anomaly detection, deep learning helps identify criminal activity and the individuals associated with it. These models analyze compound relations in the data and help law juridical predict and prevent crime, as shown in Figure. Challenges include ensuring objectivity, addressing intolerance, and maintaining privacy. The combination of deep learning with crime analytics modifies conventional approaches, enabling more dynamic, dedicated programs to struggle with and prevent crime. It helps to improve models, improve accuracy, and learn about the ethical aspects of responsible implementation of the criminal justice system. Figure 7 Survey of various research papers,

The use of deep learning models for weapon detection and facial recognition has led to highly effective implementation work. In this paper, the models can be categorized into two primary key factors: 1. Classification method 2. Object detection method. These models have shown excellent results in various studies, highlighting their reliability and efficiency in the fields of weapon detection and facial recognition.

3.3.1 Classification model

Convolution neural network

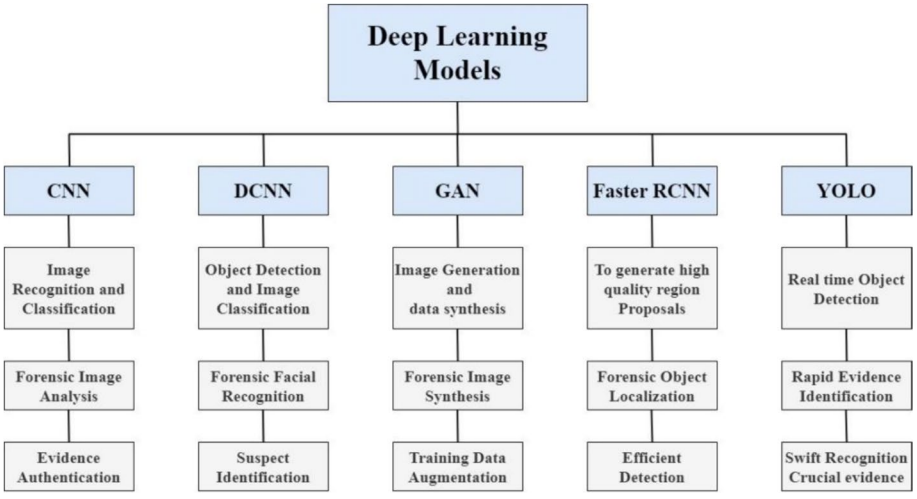


Fig. 7 Deep learning models

Convolutional neural networks (CNNs) play an important role in criminal analysis by improving the efficiency and accuracy of tasks such as weapon detection, behavior analysis, and facial recognition. Initially, the input image is input to extract features from the input image such as edges and shapes, and subsequently enters ReLU layers to establish non linearity models, which permit the study of complicated patterns, 3D convolution neural networks (CNN), and auto encoders (AEs), to compress 3D models with high resolutions [45]. The pooling layer reduces the spatial dimensions of the feature maps, and then fully connects layers are used to classify the image and video into specific categories, such as identifying a weapon or face; finally, the probability scores of each class are converted. In [12], two types of methods, standard convolution neural network (CNN) and pretrained CNN, VGG-16, VGG-9, and InceptionV3 utilized these models to easily predict criminals on the basis of facial expression. In Fig. 8 provided images described as CNN architectures used for weapon detection. Input image construction involves several weapons, and the next feature extraction step involves a convolution layer and pooling layers to help with feature mapping. The flattening layers involve the conversion of the final set of feature maps into a one-dimensional vector, and the next steps to classify the fully connected layer process the flattened vector to perform classification and the final process to predict the ability of the output layer to assist the pistol, revolver, gun, rifle, and handgun to classify the weapons very swiftly and accurately.

Deep convolution neural network

Deep convolution neural networks (DCNNs) are advanced neural network architectures that can perform well in image and video analysis. They combined multiple layers, such as convolution, rectified linear units (ReLU), pooling, and fully connected layers, which work together to automatically extract features and learn hierarchical features from the input data [46]. These networks can increase the accuracy and efficiency of automated systems in criminal analysis.

The key point surface representation (KSR) method is computationally efficient and fast for merging the scanned parts together. A deep convolutional neural network can recognize hand gestures with good performance in a wide range of gestures [47]. The evolution normalization layer normalizes [48] the spatial evolution of gestures in real time and integrates them smoothly into existing neural network architectures. Figure 9 shows the deepCNN architecture for image classification, especially for weapon detection. The process begins with an input image and advances through a feature extraction phase within a convolutional block. This block includes a convolution layer for detecting features, a ReLU activation function to introduce nonlinearity, normalization to stabilize learning, and a pooling layer to decrease spatial dimensions. The output is then flattened

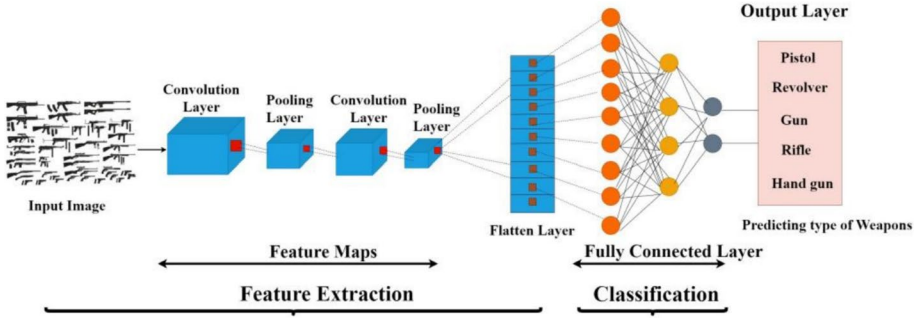


Fig. 8 Convolutional neural network architecture

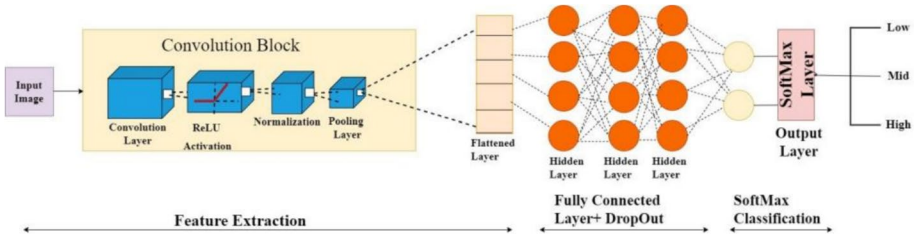


Fig. 9 Deep convolution neural network architecture

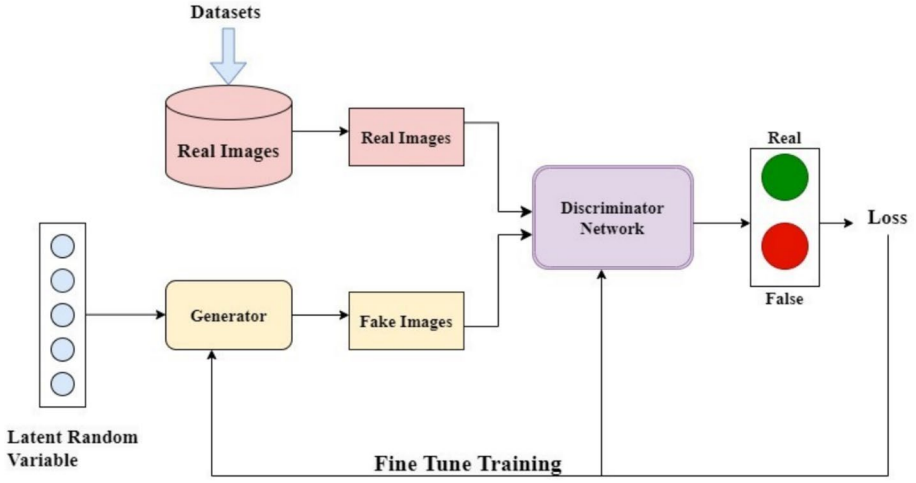


Fig. 10 Architecture of the generative adversarial network

into a single-dimensional vector for the fully connected layers. These layers consist of neurons that learn high-level features, and DropOut is applied to prevent extension. The final SoftMax layer outputs a probability distribution over class labels, ensuring that probabilities sum to one. The output layer subsequently outputs the classification result, which represents the confidence level of weapon detection.

Generative adversarial network

GANs, also known as generative adversarial networks (GANs), play a crucial role in criminal investigations by producing authentic synthetic data. This approach assists in the development of realistic scenarios for training law enforcement personnel and enhancing simulation-based training programs [27]. Consequently, it enhances their readiness and ability to make informed decisions in intricate and ever-changing situations. Moreover, GANs can anonymize sensitive data, enabling secure information sharing during collaborative invasive endeavors while safeguarding privacy. Figure 10 illustrates a generative adversarial network (GAN) framework attuned to facial recognition. With respect to the dataset of original images, the generator creates fake images from latent random variables, whereas the discriminator network evaluates both real and fake images. The discriminator network focuses on distinguishing between two signals, providing a loss signal for fine-tuning. The generator enhances the learning process to create more realistic images, increasing the difficulty of the discriminator. This adversarial training helps in generating high-quality synthetic faces, enhancing the robustness of the facial recognition model.

3.3.2 Object detection model

Faster R-CNN

Faster R-CNN is a dynamic object detection framework broadly used in criminal analysis, because of its high accuracy and efficiency. It incorporates a region proposal network (RPN) to produce regions of interest and convolutional layers to extract detailed features from images. This allows for clear identification and classification of objects, such as weapons and faces, enables real-time surveillance capabilities and enhances security measures by automating the analysis of large amounts of data, reducing false positives, and improving response times. In study J., Liang et al. [18] proposed Faster R-CNN with an atrous RPN that examines object backgrounds at different scales and slides an organization of filters that are applied to the final convolutional feature map, increasing the dilation rates.

Figure 11 represents an object detection framework using a convolution neural network (CNN), in which an input image is passed through convolutional layers to withdraw feature maps. These feature maps are fed into a regional proposal network (RPN) to identify potential object regions. The regions identified underwent region of interest (ROI) pooling for standardization of their size. Finally, a classifier processes the pooled regions to output classification labels and bounding box regressions, pinpointing object locations and types within the image.

You only look once

Law enforcement and security agencies can use YOLO for a variety of purposes, such as surveillance, monitoring congested areas, or analyzing surveillance footage. YOLO's ability to quickly identify and locate targets in real-time makes it a valuable tool in situations where timely action is critical, allowing authorities to quickly respond to potential security threats or criminal activity. The development of YOLO (you only look once) in real-time object detection has been characterized by continuous innovation and performance improvements.

YOLO v1, which introduces single-frame detection, improved the efficiency of YOLO v4 and the simplicity-focused architecture of YOLO v5; each iteration sought a balance between speed and accuracy. YOLO v8 provides immediate support for essential features such as searching, grouping, and object segmentation, with an emphasis on usability, as shown in Fig. 12. Although the YOLO series faces challenges such as small object detection and sample imbalance, it is still at the forefront of computer vision, influencing development and applications in various fields. YOLOv1 searches for objects via inversion on the basis of spatially separated bounding boxes and their class properties. This approach achieved an average precision (AP) of 63%. YOLOv2 is faster than two-stage deep-learning object detectors and has an accuracy of 93.3%. The YOLOv3

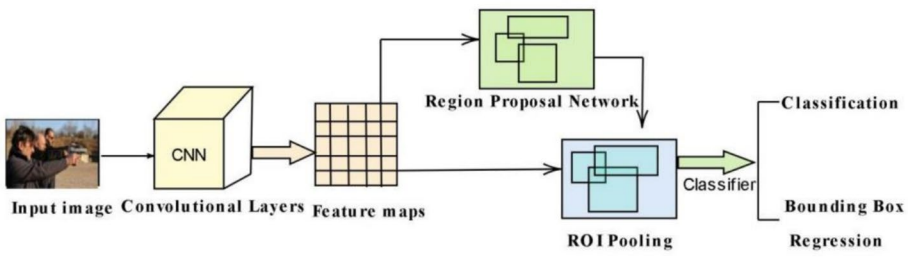


Fig. 11 Faster regions convolution neural network architecture

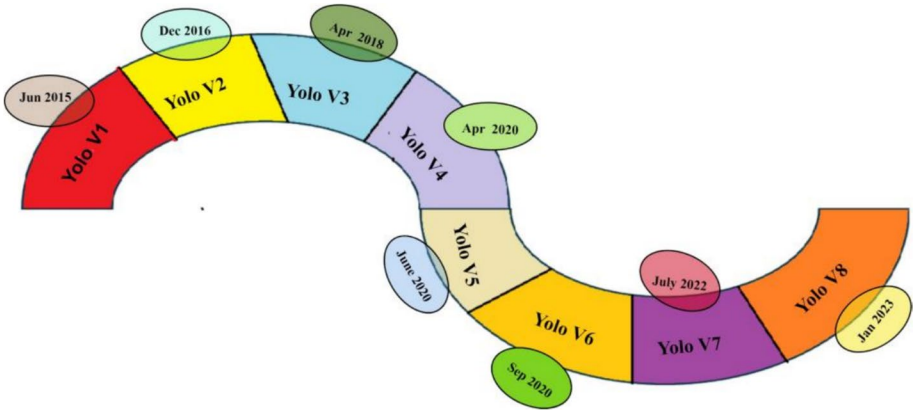


Fig.12 YOLO versions

object detection algorithm identifies specific objects in videos, live feeds, and photos. The accuracy rate was 98.87% 1. small object detection 2. high memory is needed. The YOLOv4 optimal speed and accuracy of object detection Optimal Speed and Accuracy of Object Detection achieved a high accuracy of 93.62% but with 6.1 FPS. Struggles for detecting small objects. YOLOv5 uses a new method to generate anchor boxes, called "dynamic anchor boxes". A 99.3% AP50 was achieved in 74 s. The imbalance between the positive and negative samples in YOLOv6 involves identifying the location of an object (or multiple objects) in the image frame. This approach achieves a 93.48% improvement in the range of models and continuously improves the detection performance. YOLOv7 can be used to track materials as they move through a production line. It is 95% accurate and cannot detect cars, motorcycles, or persons who are very far off or small. YOLOv8 includes out-of-the-box support for searching, grouping, and object segmentation functions and achieves a mean average precision (mAP) of 45.6 on the validation set.

YOLO frameworks are usually preferred over deep learning YOLO concepts because they save time, memory, and resources such as CPUs and processors. YOLO frameworks can yield more accurate results than machine learning models constructed from scratch; this approach is also known as transfer learning. In this case, the dark net models in YOLO are trained on a customized dataset of guns created via Roboflow. Roboflow is a developer tool that helps you build computer vision models quickly and accurately for better results. YOLOv8 is a state-of-the-art (SOTA) model that follows the footsteps of previous generations, adding new functionalities and enhancements for improved performance, scalability, and productivity. It supports various AI vision functions, such as detection segmentation, tracking, estimation, and classification.

Hyperparameter Tuning: It used Ultralytics built-in hyperparameter tuning framework for YOLO models with implemented genetic algorithms to fine tune parameters including learning rate, momentum, and weight decay. For CNN models we performed grid search optimization of: learning rate, dropout rate, batch size, and the number of hidden layers.

Cross-Validation: To make sure our model was robust and generalizable across data splits we used K-Fold Cross-Validation (traditionally with $k = 5$).

Data Augmentation Techniques: To increase diversity in the training data, we used Mosaic Augmentation, random flips, scales, translations, tweaks in HSV, and random

erasing to augment the data. All augmented data was realized using Albumentations and our own custom pre-processing pipelines.

4 Datasets

Datasets are crucial for collecting weapon detection data because they provide the necessary training data for deep learning algorithms to recognize and identify weapons accurately. High-quality datasets ensure that models can generalize well to real-world scenarios, improving detection rates and reducing false positives.

4.1 COCO datasets

The Common Objects in Context (COCO) datasets include large-volume scale object detection, segmentation, and captioning data. COCO contains more than 330,000 images with more than 200,000 labeled images, involving 80 object categories and 1.5 million object instances [10]. Coco features are enriched in annotations and include object detection, such as bounding boxes, segmentation (masks), and key points for human poses and image captions, making them comprehensive resources for several computer vision tasks.

4.2 Hockey fight videos

Hockey fight video datasets are formed as video clips capturing physical arguments; these datasets are utilized to train and evaluate models in action recognition and analysis [14]. The features of this dataset typically include annotations such as the beginning and end of fights and contextual information.

4.3 Movies fight detection datasets

The Movies Fight Detection Dataset includes video clips from various films featuring fight scenes annotated for the presence of physical combat [14]. It is used for developing and evaluating models in action detection and video analysis tasks. The features of the movies fight detection dataset include temporal annotation, spatial annotation, contextual metadata and action labels.

4.4 Crowd activity datasets

The crowd activity datasets comprise video recordings of several crowd behaviors and activities and are used for training and evaluating crowd behavior analysis and anomaly detection models [14]. The datasets includes diverse scenarios such as gatherings, protests, sporting events, and pedestrian movements. Crowd management benefits safety monitoring, event management, anomaly detection, and urban planning.

4.5 Wider face datasets

The wider face dataset is a standard dataset for facial detection that includes 32,203 images and 393,703 labeled faces with a high degree of variability in scale, human pose, and obstruction. It is commonly used for training and evaluating facial detection algorithms [21]. The features include diverse annotations, subsets, contextual varieties, and standardized splits. The use of wider facial datasets achieves the benefits of robust model training, bench marking, improved accuracy, and application versatility.

4.6 Internet movie firearms

The Internet Movie Firearms Database (IMFDB) dataset consists of detailed datasets about firearms used in movies and television shows [20]. This approach involves narratives, appearances, and technical details about the weapons featured in various media platforms. The features included comprehensive entries, visual references, media metadata, and categorization.

4.7 CCTV datasets

The CCTV datasets consist of video footage from surveillance cameras and are used for training and evaluating models for security, object detection, and activity recognition. These methods often include annotations for various activities, such as pedestrian movement, vehicle tracking, and suspicious behavior [12]. Utilized CCTV datasets Accurate detection, contextual understanding, enhanced analysis, real-time monitoring, and post incident review were performed on the CCTV datasets.

4.8 Photograph and video datasets

Photograph and video datasets are collections of several images and video clips used for different tasks in computer vision and machine learning, such as object detection, image classification, and action recognition [49] and deep learning help with automated detection and classification utilizing photos and sensor data [50].The photograph datasets included high-resolution images, annotations, diverse content and metadata, video datasets such as continuous clips, temporal annotations, object tracking and action recognition. The datasets support the real-time application and behavior analysis.

4.9 Robbery CCTV videos

Robbery CCTV videos are surveillance video recording clips that capture instances of robbery or attempted robbery [20]. These videos are typically used for testing and evaluating models in security, crime detection, and forensic analysis.

4.10 Google image datasets

These datasets refer to collections of various images from Google source images [27] that are utilized in various deep learning and computer vision tasks. A wide range of categories involves images from numerous categories, such as people and objects. These datasets help augment existing datasets with diverse and representative images to improve model performance.

4.11 Real-world masked face recognition datasets (RMFRD)

Specialized datasets are designed to address the challenges of recognizing faces that are partially obscured by masks [19]. The datasets are divided into three scales: small-scale datasets, which range from 5000 to 10,000 images; medium-scale datasets range from 30,000 to d100,000 images; and large -scale datasets, which range from 200,000 to 1,000,000 images.

4.12 NIST mugshot datasets

Datasets collection of Mugshot images used primarily for biometric research, including facial recognition and forensic identification. These datasets were generated by the

National Institute of Standards and Technology (NIST) and utilized to develop and evaluate algorithms for facial recognition and other biometric technologies [17]. For example, NIST Special Database 24 largely uses a datasets of more than 10,000 mugshot images from a variety of individuals, and NIST special database 27 contains images from multiple sources, including mugshots for broader biometric research applications.

The datasets used in this research were all publicly available datasets that were designed for detecting faces and weapons. COCO for general object detection, WIDER FACE and NIST Mugshot datasets for facial recognition, IMFDB and CCTV footage for weapon detection. There were Hockey Fight and Crowd Activity datasets to support violence detection. Image data for facial recognition was sourced from online images via Google images, to add diversity. The source of the dataset location and datasets can be found in Table 4 for reference and in order to enable researchers with reproducibility.

In presented Tables 3 and 4 summarize the achievements of the models on several datasets for tasks such as object detection, facial detection, and violence detection. Notably, a faster RCNN with ResNET 50 achieved an average accuracy of 62.11% on the gun and face datasets. The MobileNet V-2 and Xception datasets have high accuracies of 92.2% and 91.6% respectively for certain operations. In the field of hockey match detection, 96.94% accuracy is achieved, while the CCTV dataset yields a significant improvement in detection speed from 10 to 16 frames per second (FPS). The accuracy of masked face detection in the real world is 91.3%, and YOLOv7 outperforms YOLOv6

Table 3 Datasets for crime detection through deep learning methods

S.No	Refs.	Algorithm	Datasets	Testing	Performance
1	Akshaya [21]	YOLOv4	Internet Movie Firearms, WIDER FACE datasets	Faster RCNN Resnet 50:1(mAP)	Accuracy efficient Det: B0 (0.06) seconds Than the ensemble scheme ENSD 3(0.17)seconds
2	Chatterjee[10]	Greedy Ensemble Object Detection	COCO datasets	10epoch-71%	Accuracy 98%
3	SNikkath Bushra[14]	YOLO v5	Hockey Fight Detection Datasets, Academic Tor rents	Accuracy 96.94% for validation Accuracy 80%	98.3 ± 0.81%
4	Apar Jaggi[12]	Fuzzy K-means Clustering algorithm	CCTV Datasets	Threshold is 0.5	Detection speed from approximately 10 FPS to about16 FPS
5	Sanika Tanmay[49]	Haar Cascades DNN	Photographs and videos	Mobile Net V2:98.6%	Accuracy 98.55%
6	Rebello[20]	Deep Fake Detection	Internet Movie Firearms database Robbery CCTV Videos	Datasets 1:260 Datasets 2:784 Datasets 3:999	Accuracy:91.73%
7	E Poornima [27]	GAN	Google images	88.5%	Accuracy
8	Khalid Raihan [15]	YOLOv7	CCTV videos from YouTube, real- world images of the weapons from Kaggle, and Open Images Dataset's	Short guns:4370 long guns:3426 knives:1984	Accuracy
9	Swati Jagtap [19]	Mtcnn algorithm	Real-world Masked Face Recognition Datasets(RMFRD)	47.2%	Accuracy of 91.3%(RMFRD)
10	Yesun Utomo [17]	YOLOv8	NIST mugshot datasets	97.183%	Accuracy 97%

Table 4 Datasets with source link

S.No	Datasets	Source link
1	Coco datasets	http://surl.li/rukqno
2	Hockey fight videos	http://surl.li/bqdmke
3	Movies Fight Detection	http://surl.li/nwlpxr
4	Crowd Activity Datasets	http://surl.li/vhvfmc
5	Wider face datasets	http://surl.li/aoxzqv
6	Internet movie firearms	http://surl.li/mewxwu
7	CCTV datasets	http://surl.li/vabiig
8	Photograph and video datasets	http://surl.li/svebcc
9	Robbery CCTV videos	http://surl.li/xicucr
10	Google image datasets	http://surl.li/gbiiiz

Crime Types / Datasets	Robbery	Violence	Assault	Burglary	Forgery
Internet Movie Firearms	✓	✓			
Database Robbery CCTV Videos	✓				
Wider Face dataset's			✓		✓
CCTV Datasets	✓	✓	✓	✓	✓
NIST Mugshot				✓	
MISD dataset					✓
COCO dataset	✓			✓	
Hockey Fight Detection				✓	
Academic Torrents Crowd Violence		✓			
Google Images		✓	✓		

Fig. 13 Dataset analysis

and YOLOv5 with 96% accuracy, This analysis provides insight into the performance of different deep learning models across multiple datasets and applications. This table lists several datasets that are valuable for deep learning related to weapons detection, crowd activity analysis, and facial recognition. It includes links to listed datasets covering a range of sources, including Kaggle, academic torrents, and specialized collections such as the wider face datasets and CCTV footage from YouTube, providing a diverse foundation for developing and testing models in these domains. This paper addresses the challenges of detecting weapons and recognizing faces via the above-mentioned datasets for different crime types and describes how deep learning can achieve accurate results. Figure 13 shows, that Internet Movie Firearms extract frames where weapons are used, providing a rich source of labeled weapon imagery and focusing on each segment where weapons are involved in robberies. wider face datasets are used for high-quality facial images under various conditions to enhance facial recognition models. CCTV datasets are used to detect suspicious behavior and weapon use in real-world settings. MISD datasets focus on specific forgery-related images and help to predict fraudulent

activities. COCO datasets for detecting common objects and scene contexts. Hockey fight detection frames of physical alterations are used to train models for detecting violent behavior. Academic torrents crowd violence datasets to understand crowd dynamics and detect violence. Google Images scrapes more images of faces and weapons to augment training datasets.

Weapon detection challenges in performing weapon detection on datasets used in data augmentation to improve the datasets through several augmentations such as rotation, scaling, and noise addition, to make the model robust. Transfer learning to utilize pre trained models such as YOLO or Faster RCNN were trained on COCO datasets to detect objects accurately and fine-tuned with specific weapon imagery from internet Movie Firearms and the Database Robbery CCTV Videos. Multiscale detection implements feature maps to detect weapons of different sizes and resolutions, especially in CCTV footage and weapon detection datasets, and works related to trained models and the detection of certain crime types. Figure 14 shows how each datasets contributes to several aspects of crime detection and analysis, emphasizing their specific applications and strengths.

These charts illustrate various weapon detection datasets and provide a clear and precise overview of how each dataset is used for specific tasks, helping in understanding the roles of each dataset in comprehensive crime detection strategies.

Datasets for facial recognition

High-quality annotations were generated using wider facial datasets and NIST Mugshot files high-quality facial annotations under different conditions. Facial matching and verification to employ advanced developed models such as Arc Face or VGGFace2 for training, leveraging diverse facial data from these datasets. Occlusion Handling trains with augmented data where faces are partially occluded to increase the degree of model error in real-world scenarios. Figure 15, illustrates the use of various facial recognition datasets, such as the wider Face, CCTV, NIST Mugshot, and Google Images datasets.

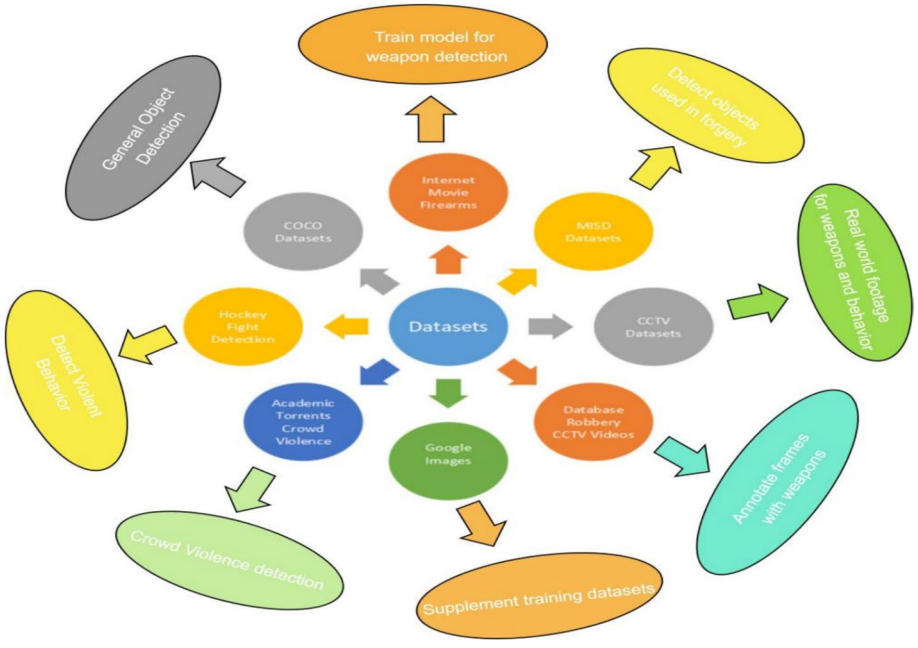


Fig. 14 Dataset usage for weapon detection

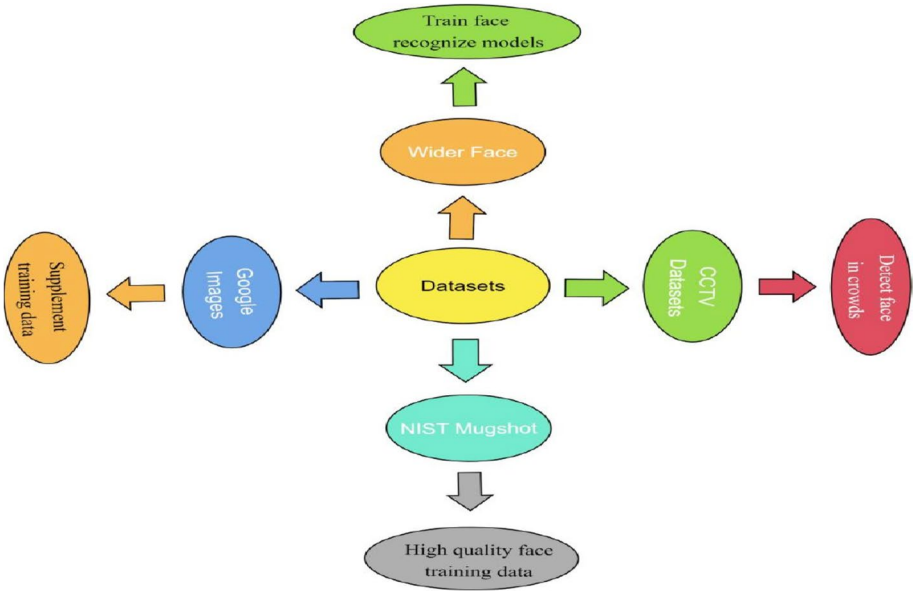


Fig. 15 Dataset usage for facial recognition

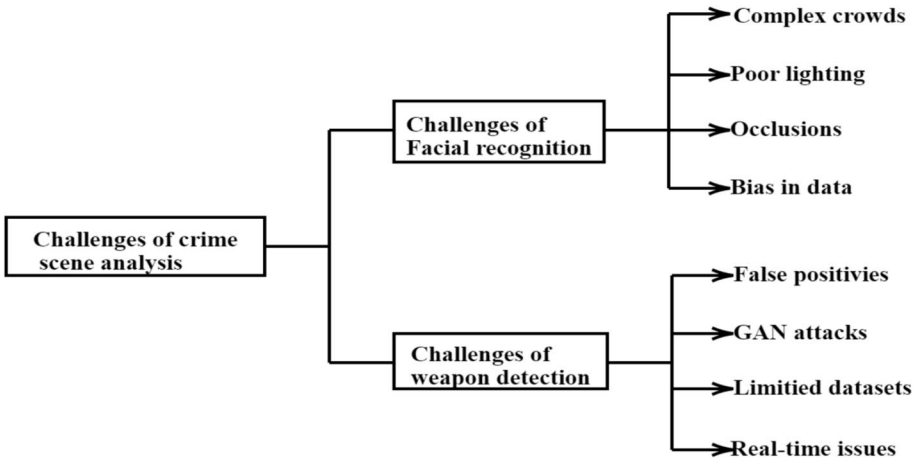


Fig. 16 challenges of detection

This bar is labelled for its specific usage, such as training facial recognition models or detecting faces in crowds. This visualization graph shows the application of different datasets in facial recognition tasks.

5 Challenges

In recent years, detecting crime activities has presented terrifying challenges, and the primary issue lies in the scarcity of high-quality, annotated datasets due to privacy concerns and limited access, which hampers effective model training. Advanced architectures such as YOLO or Faster R-CNN are overwhelmed by the computational demands of real-time crime detection. The inability to detect crimes or mistakes among criminals is a major problem that persists in real-time crime detection. Additionally, the trustworthiness and interpretability of black-box deep learning models that use CNNs and RNNs complicate the process as shown in Fig. 16.

5.1 Challenges for weapon detection

CNNs for weapon detection have been successful in this domain but require large annotated datasets of weapons which are scarce and difficult to obtain because of privacy concerns. These methods excel at sequential data analysis but need an understanding of temporal patterns and context when applied to weapon detection, such as when distinguishing between normal activities and potential threats. GANs present both opportunities and challenges in weapon detection by obtaining fake images of weapons, which could confound CNN-based detection systems. This susceptibility approach involves robust training methods to mitigate adversarial attacks and ensure reliable detection accuracy. Overcoming these challenges requires enhancement in datasets curation and model error-freeness against adversarial inputs, among interpretable trustworthy weapon detection systems.

5.2 Challenges to facial recognition

The use of classification and detection models for facial recognition presents a number of significant hurdles. Classification models are efficient at identifying faces under controlled settings; however, they lack robustness to variations in lighting, facial expressions, and occlusions which can have significant accuracy implications. This requires substantial computational resources as well as efficient processing techniques to enable these models to scale for vast datasets or real-time applications. Moreover, there is always a chance of bias within training data resulting from various acts across various analytical groups thereby increasing issues about fairness and opportunity. Object detection methods such as YOLO and Faster R-CNN are excellent at detecting rapidly moving objects in real time but may have insufficient specificity for accurate facial recognition, particularly in crowded or complicated scenes. Furthermore, since the interpretability of these models concerning recognizing individual identities or distinguishing between similar faces is still an issue, their reliability in security issues is compromised. These challenges can be addressed through improvements in modern robustness, dataset diversity, and privacy-preserving techniques. Importantly, ethical guidelines covering transparency practices suggest ways to improve the interpretability and fairness of such technologies. Therefore trust can be developed around their application on multiple fronts, from security and surveillance to personalized services and beyond.

6 Results and discussion

Crime analysis involves the use of data visualization to provide a complete view of the assessment and types of criminal activities, such as robbery, assault, violence, forgery, and burglary. Spatial or temporal analysis helps to examine law enforcement and police departments and develop targeted crime prevention strategies. In this paper, visual tools such as pie charts, heat maps, and graphs improve understanding and support effective responses to social challenges and improve public safety. Through visual representation, it provides an exhaustive review of the spread and composition of several criminal activities. This pie chart shows the results of the two models CNN and YOLO, indicating their common application and reliability in object detection tasks. SSD, R-CNN, and Fast RCNN also exhibit considerable segregation, indicating their success in specific scenarios. LSTM and the GAN occupy smaller segments. Efficient Det and Faster R-CNN advanced the capabilities and provided good results in high-performance contexts. This

distribution shows the various ranges of models used to improve crime detection and analysis. Figure 17 shows how to understand patterns and trends in crime occurrence, which facilitates informed decisions by law enforcement agencies and policymakers. By presenting the types of crime in a visually accessible form, this research increases the understanding and support of targeted crime prevention and intervention strategies. Finally, imaging is a valuable tool for responding to social challenges and improving public safety. The pie chart illustrates the proportion of the dynamic status among several models, including the CNN, LSTM, GAN, RBFN, and YOLO.SSD, R-CNN, fast R-CNN, faster R-CNN, and efficient detection.

Figure 18 the graph shows the numerical scores for different crime types (robbery, violence, assault)for different models, including the CNN, YOLO, SSD, R-CNN, Fast R-CNN, Faster R-CNN, and Efficient Det, burglary, forgery models. In the domains of weapon detection and facial recognition, the use of state-of-the-art deep learning models is important for ensuring high accuracy and reliability. Convolutional neural networks (CNNs) exhibit exceptional performance with an accuracy of 98% [11], making them highly effective for recognizing intricate patterns in images, which is crucial for both weapon detection and face recognition. In [20] recurrent neural networks (RNNs), which achieve an accuracy of 81%, are useful for sequence prediction and temporal data analysis, making them valuable in video surveillance where tracking movement over time is essential. Long short-term memory(LSTM) networks, a type of RNN, have an accuracy of 77.31% [15] and are particularly adept at maintaining context over time,

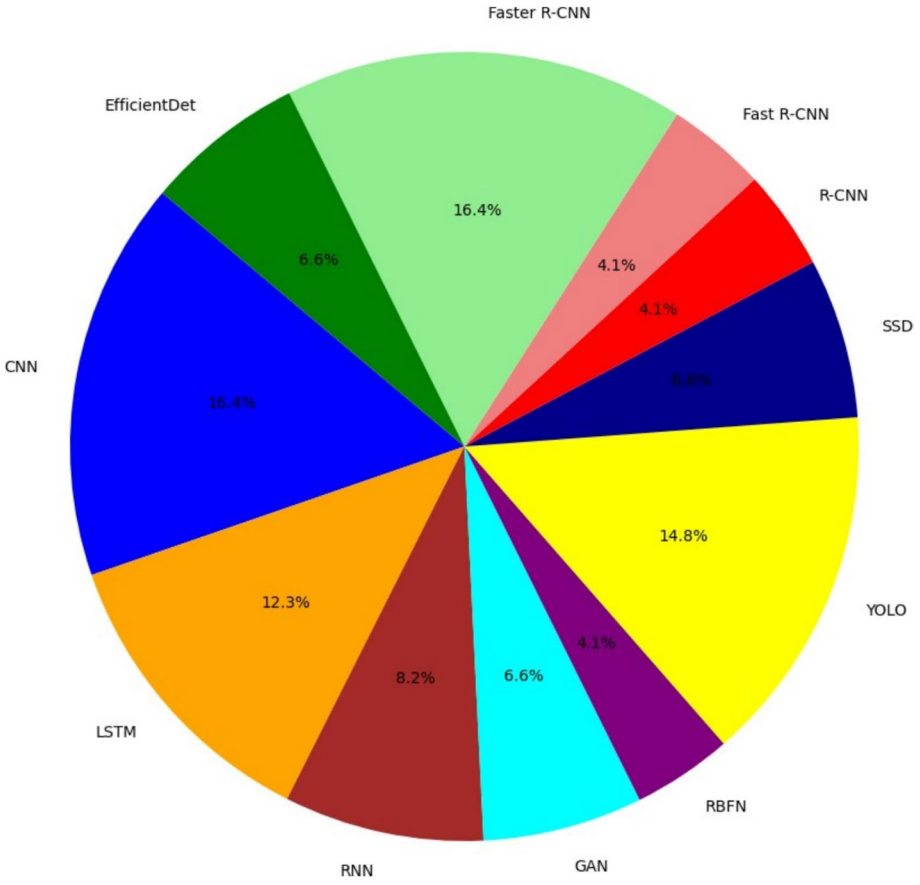


Fig. 17 Distribution of model status

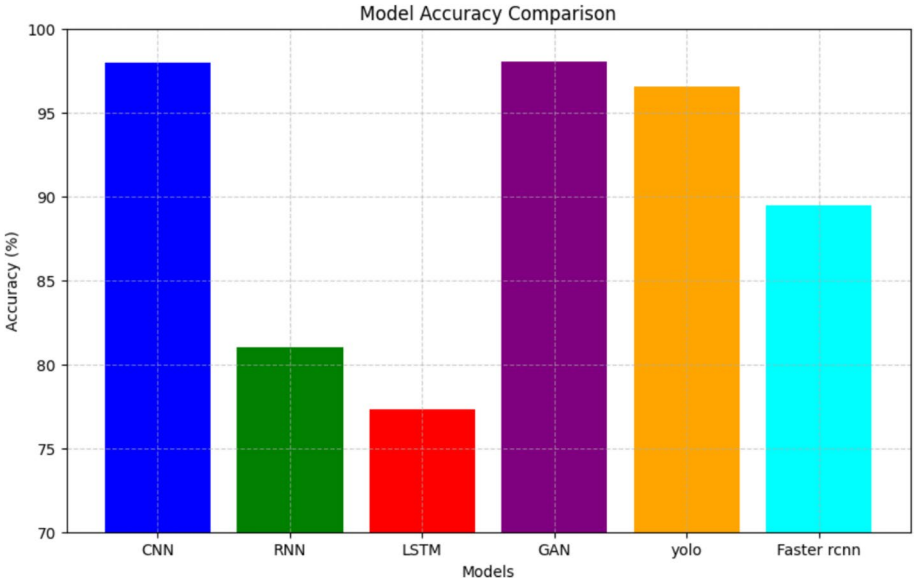


Fig. 18 Comparison of Model Accuracy

aiding in the detection of ongoing suspicious activities. Generative adversarial networks (GANs) [51], which have a high accuracy of 98.02%, are powerful tools for generating synthetic data to improve training datasets, thereby improving the robustness of detection systems. The YOLO model [14], known for its instantaneous object detection capacity, achieves an accuracy of 96.53%, making it a prominent choice for real-time surveillance and immediate region proposal networks with CNNs to achieve high detection accuracy, and is particularly useful for the precise localization of faces or weapons in complex environments. In addition to accuracy, we evaluated the models using precision, recall, F1-score, and AUC. These metrics provide a deeper insight into the model's true positive rate, balance between precision and recall, and overall classification robustness in imbalanced crime datasets.

Faster RCNNs have an accuracy of 89.47% [52], together these models form a comprehensive and effective suite for enhancing security through advanced image and video analysis technologies as explained in systematic review Fig. 19 the analysis of various crime detection models, such as CNN, YOLO, EfficientDet, and Faster R-CNN, reveals distinct patterns of effectiveness across different crime categories. For robbery and assault, CNN, YOLO, and EfficientDet emerge as the leading models, with CNN being slightly better at detecting robbery, suggesting that these models are robust at detecting such incidents. In the real-time suitability assessment, we assessed inference times and memory footprints between the models. YOLOv5 and YOLOv7 had a higher inference rate (~ 10–16 FPS) and used less memory and are therefore applicable for live CCTV streams. Faster R-CNN had a higher accuracy level at localising an object, but higher latency and memory footprints meant less real-time suitability. The EfficientDet model was an appropriate compromise with reasonable inference speed with high accuracy, and therefore a good model for edge devices.

Forgery detection, however, presents an even distribution effectiveness among EfficientDet, CNN, and YOLO, indicating that no single model has a significant advantage, allowing for interchangeable use among these models. In violence detection, CNN and YOLO show equal efficacy and offer flexibility in model choice. Burglary detection

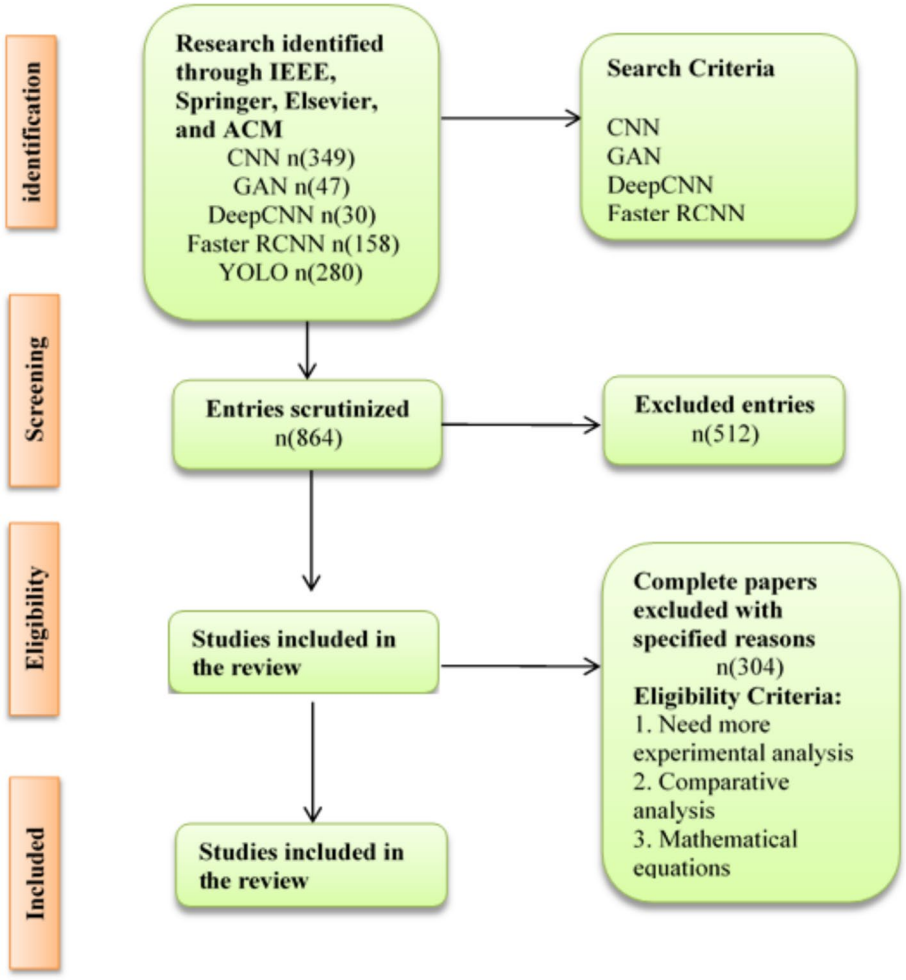


Fig. 19 Systematic Review

approaches Faster R-CNN and YOLO, highlighting their superior performance in this domain.

7 Conclusion

The examination of crime distribution reveals the intricate and multifaceted nature of criminal activities. The review emphasizes the importance of facial recognition and weapon detection technologies for public safety and proactive monitoring. The deep learning models, including Convolutional Neural Networks (CNNs) and YOLO, can identify and classify crime events in a variety of contexts. The models can be applied to real-world monitoring situations. Both the CNNs and YOLO may provide the capability to assess risk in 'live' monitoring situations and these technology has the potential to detect and identify threats proactively; however, there are more practical issues such as demands for varying light and weather conditions, minimizing false-positives, and ensuring the speed necessary for responding to opportunities as they present in real-time monitoring situations. Future work should focus on optimizing these models for greater generalized use in real-world settings, including engaging sophisticated data augmentation methodologies, and using transfer learning to account for different crime patterns with limited retraining. Moreover, issues of ethics, including data privacy

issues and preventing bias, need to be addressed even while understanding the need and right to use AI, with trust in accountable organizations, responsibly in surveillance. The inclusion of additional performance metrics such as precision, recall, F1-score, and AUC enhances the reliability of the evaluation and better reflects the model's real-world effectiveness in crime detection scenarios. Overall, it is clear this study has set the groundwork for evolving intelligent crime prevention through deep learning, but innovation and evaluation methods continue to be developed to ecological applied contexts. Further work should be completed on the practical merging of these deep learning techniques into working surveillance systems. There is also a need for improving accuracy, reducing false positives and improving real-time processing.

Author contributions
S.P. Developed the review protocol, conducted the database searches and study screening following the PRISMA guidelines, performed data extraction and quality assessment, synthesized the findings, and drafted the manuscript. M. V Guided the design and refinement of the review protocol, supervised the screening and synthesis process, ensured compliance with PRISMA reporting standards, provided critical feedback on analysis, and revised the manuscript for important intellectual content. Both authors approved the final manuscript and agree to be accountable for all aspects of the work.

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Data availability
No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate
Not applicable.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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References

1. Huang K-T, Lin H-L, Zheng Q-M. Discussion on the safety protection measures of UAV campus. IEEE; 2019. pp. 304–5.
2. Ou J. Classification algorithms research on facial expression recognition. Phys Procedia. 2012;25:1241–4. https://doi.org/10.1016/j.phpro.2012.03.227.
3. Pantic M, Rothkrantz LJM. Expert system for automatic analysis of facial expressions.
4. Nurzyńska K, Smołka B. Facial displays description schemas for smiling vs. neutral emotion recognition. In: Lecture Notes in Artificial Intelligence (sub series of Lecture Notes in Computer Science) vol. 9119. 2015. pp. 594–605.
5. Cohn J. Advances in behavioral science using automated facial image analysis and synthesis. IEEE Signal Process Mag. 2010;27:128–33. https://doi.org/10.1109/MSP.2010.938102.
6. Kumar N, Jaglan V, Agrawal A. A review of various face prediction models using image processing. Int J Computer Appl. 2016;142:18–22. https://doi.org/10.5120/ijca2016909714.
7. Bhatti MT, Khan MG, Aslam M, Fiaz MJ. Weapon detection in real-time cctv videos using deep learning. IEEE Access. 2021;9:34366–82.
8. Tiwari RK, Verma GK. A computer vision based framework for visual gun detection using harris interest point detector. Proced Computer Sci. 2015;54:703–12.
9. Haji S, Varol A. Real time face recognition system (rtfrs). IEEE; 2016. pp. 107–11.
10. Chatterjee R, Chatterjee A, Pradhan MR, Acharya B, Choudhury T. A deep learning-based efficient firearms monitoring technique for building secure smart cities. IEEE Access. 2023;11:37515–24.
11. Pavithra S, Muruganantham B. Dl based automatic robbery detection using video surveillance in residential areas. IEEE; 2022. pp. 1–7.
12. Jaggi A, Aggarwal A, Gupta A. Identifying anti-social activities in surveillance monitoring applications using deep-CNN based algorithms. IEEE; 2023. pp. 1–5.
13. Apon TS, Chowdhury MI, Reza MZ, Datta A, Hasan ST, Alam MGR. Real time action recognition from video footage. IEEE; 2021. pp. 1–6.
14. Bushra SN, Sibi SA, VijayaKumar K, Niveditha M. Predicting anomalous and consigning apprise during heists. IEEE; 2023. pp. 1–8.
15. Talha KR, Bandapadya K, Khan MM. Violence detection using computer vision approaches. IEEE; 2022. pp. 544–50.

16. Hu X, Wei Z, Zhou W. A video streaming vehicle detection algorithm based on yolov4 5. IEEE; 2021. pp. 2081–6.
17. Utomo Y, Kusuma GP. Masked face recognition: progress, dataset, and dataset generation. IEEE; 2021. pp. 1–4.
18. Rehman A, Fahad LG. Real-time detection of knives and firearms using deep learning. IEEE; 2022. pp. 1–6.
19. Jagtap S, Chopade NB, Tungar S. An investigation of face recognition system for criminal identification in surveillance video. IEEE; 2022. pp. 1–5.
20. Rebello L, Tuscano L, Shah Y, Solomon A, Shrivastava V. Detection of deepfake video using deep learning and MesoNet. IEEE; 2023. pp. 1022–6.
21. Akshaya P, Reddy PB, Panuganti P, Gurusai P, Subhahan A. Automatic weapon detection using deep learning. IEEE; 2023. pp. 1–7.
22. Shah SAA, Al-Khasawneh MA, Uddin MI. Review of weapon detection tech- niques within the scope of street-crimes. IEEE; 2021. pp. 26–37.
23. Shanthi P, Manjula V. "38 Integrated Deep Learning Framework for Weapon—Behavior Analysis." Power Energy and Secure Smart Technologies (2025):27.
24. Kumar, BHM, et al. Advanced meta-heuristic algorithm based on AlBiruni earth radius and particle swarm optimization methods for oral cancer detection. In: 2024 International Conference on Knowledge Engineering and Communication Systems (ICKECS). Vol. 1. IEEE; 2024.
25. El-Kenawy E-S, et al. Greylag goose optimization: nature-inspired optimization algorithm. Expert Syst Appl. 2024;238:122147.
26. Alkanhel R, et al. Network intrusion detection based on feature selection and hybrid metaheuristic optimization. Comput Mater Contin. 2023. https://doi.org/10.32604/cmc.2023.033273.
27. Poornima E, Pragisha K. Face sketch recognition using image synthesis and deep learning. IEEE; 2023. pp. 488–93.
28. Shanthi P, Manjula V. "Weapon detection with FMR-CNN and YOLOv8 for enhanced crime prevention and security." Scientific Reports (2025);15(1).
29. Vuyyuru LR, et al. Advancing automated street crime detection: a drone-based system integrating CNN models and enhanced feature selection techniques. Int J Mach Learn Cybern. 2025;16(2):959–81.
30. Ramzan M, et al. A review on state-of-the-art violence detection techniques. IEEE Access. 2019;7:107560–75.
31. Li M, et al. Ao-detr: anti-overlapping detr for x-ray prohibited items detection. In: IEEE Transactions on Neural Networks and Learning Systems. 2024.
32. Fang X, Liu J, Chen Y, Han J, Ren K, Chen G. Nowhere to hide: detecting live video forgery via vision-wifi silhouette correspondence. IEEE; 2023. pp. 1–10.
33. An H, Kim J, Sheng B, Li P, Jung Y. A transfer function optimization using visual salience for region of interest-based direct volume rendering. Displays. 2023;80:102531.
34. Sheng B, Li P, Fu H, Ma L, Wu E. Efficient non-incremental constructive solid geometry evaluation for triangular meshes. Graph Models. 2018;97:1–16.
35. Sreenu G, Durai S. Violence detection in crowd videos using nuanced facial expression analysis. Syst Soft Comput. 2024. 200104.
36. Aouaidjia K, Sheng B, Li P, Kim J, Feng DD. Efficient body motion quantification and similarity evaluation using 3-d joints skeleton coordinates. IEEE Trans Syst Man Cybern Syst. 2019;51(5):2774–88.
37. Wei J, Hu G, Yang X, Luu AT, Dong Y. Learning facial expression and body gesture visual information for video emotion recognition. Expert Syst Appl. 2024;237:121419.
38. Teoh K, Ismail R, Naziri S, Hussin R, Isa M, Basir M. Face recognition and identification using deep learning approach. J Phys Conf Ser. 2021;1755(1):012006.
39. Xu N, Feng W. Metafake: few-shot face forgery detection with meta learning. 2023. pp. 151–6.
40. Pangestu G, Maulana FI, Puspitasari C, Sidharta S, Sano AV, Sukardi AD Head movement detection using deep learning and face edge detection (fed) method. 2021. pp. 315–20.
41. Jiang J, Sheng B, Li P, Ma L, Tong X, Wu E. Real-time hair simulation with heptadiagonal decomposition on mass spring system. Graph Model. 2020;111:101077.
42. Karambakhsh A, Sheng B, Li P, Li H, Kim J, Jung Y, et al. Sparsevoxnet. 3-d object recognition with sparsely aggregation of 3-d dense blocks. IEEE Trans Neural Netw Learn Syst. 2022;35(1):532–46.
43. Sheng B, Liu B, Li P, Fu H, Ma L, Wu E. Accelerated robust boolean operations based on hybrid representations. Comput Aided Geom Des. 2018;62:133–53.
44. Yao J, Chen J, Niu L, Sheng B. Scene-aware human pose generation using transformer. 2023. pp. 2847–55.
45. Ertugrul E, Zhang H, Zhu F, Lu P, Li P, Sheng B, et al. Embedding 3d models in offline physical environments. Comput Anim Virtual Worlds. 2020;31(4–5):1959.
46. Kim J, Yun S-S, Kang B-N, Kim D, Choi J. Reliable multi-person identification using dcnn-based face recognition algorithm and scale-ratio method. IEEE; 2017. pp. 97–101.
47. Karambakhsh A, Kamel A, Sheng B, Li P, Yang P, Feng DD. Deep ges- ture interaction for augmented anatomy learning. Int J Inf Manage. 2019;45:328–36.
48. Zeghoud S, Ali SG, Ertugrul E, Kamel A, Sheng B, Li P, et al. Real-time spatial normalization for dynamic gesture classification. Vis Comput. 2022. https://doi.org/10.1007/s00371-021-02229-9.
49. Ratnaparkhi ST, Singh P, Tandasi A, Sindhwani N. Comparative analysis of classifiers for criminal identification system using face recognition. IEEE; 2021. pp. 1–6.
50. Ajagbe, Sunday Adeola, and Matthew O. Adigun. "Deep learning techniques for detection and prediction of pandemic diseases: a systematic literature review." Multimedia Tools and Applications (2024);83.2:5893–5927.
51. Zhang T, Tang K. An efficacious method for facial expression recognition: Gan erased facial feature network (ge2fn). 2021. pp. 417–22.
52. Bandong S, Nazaruddin YY, Joelianto E. Faster rcnn mixed-integer optimization with weighted cost function for container detection in port automation. Heliyon. 2023. https://doi.org/10.1016/j.heliyon.2023.e13213.

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